

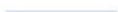
Data Cleaning with Pandas

Create a defensible path from messy observations to analytical evidence.



Learning outcomes

01  Profile missingness, duplicates, types, and invalid ranges.

02  Apply explicit rules without hiding data loss.

03  Validate assumptions before and after transformation.

The concept system

01

data quality

02

missing values

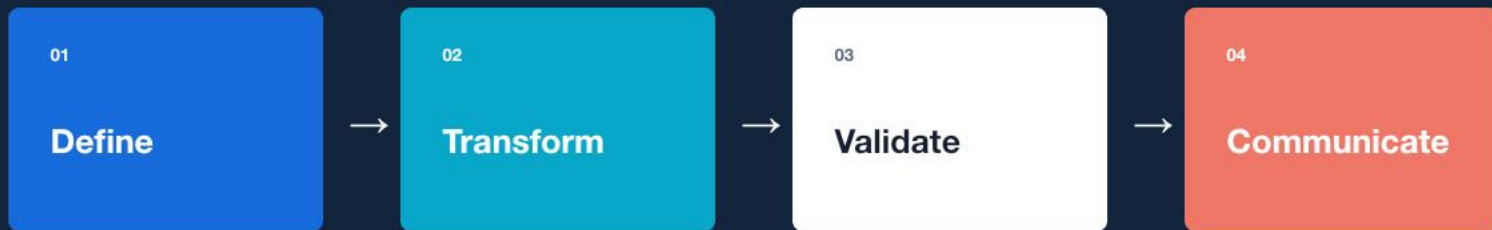
03

types

04

validation

A repeatable reasoning loop



Clean a noisy customers and orders dataset with an auditable rule log.

Guided lab

LAB BRIEF

Clean a noisy customers and orders dataset with an auditable rule log.

workbook / tests / evidence

01

Frame

Define inputs, outputs, and success criteria.

02

Build

Implement the smallest correct workflow.

03

Verify

Test normal, boundary, and failure cases.

04

Explain

Record the evidence and remaining limits.

Portfolio deliverable

A cleaning pipeline, validation report, and rejected-record table.

QUALITY GATE

-  **Reproducible**
-  **Tested**
-  **Decision-relevant**

WEEKLY ASSESSMENT

Evidence-First EDA

Cleaning 30%, reasoning 30%,
visual evidence 20%, limitations
10%, reproducibility 10%.

What should remain after the lesson?

CONCEPT

data quality + missing values

PRACTICE

Clean a noisy customers and orders dataset with an auditable rule log.

EVIDENCE

A cleaning pipeline, validation report, and rejected-record table.
